02. Dataset Understanding

Inspect the actual CSV, columns, class distribution and saved artifacts.

Project: Child Malnutrition Status Classification (ICFDR ML Project)

Source archive: ICFDR---ML-Project-main.zip | Main notebook: MalnutritionDetect.ipynb | Additional notebooks: SMOTE.ipynb and Oversampling on the whole dataset.ipynb

Dataset dimensions

The supplied Malnutrition data.csv contains 837 rows and 8 columns, with no missing values in the inspected CSV.

Columns

Column	Type / interpretation
Sex	Binary-coded sex feature (0/1 in examples)
Age	Age value; unit is not documented in the archive
Height	Height measurement; unit is not documented in the archive
Weight	Weight measurement; unit is not documented in the archive
Low Income	Income-group indicator
Lower Middle Income	Income-group indicator
Upper Middle Income	Income-group indicator
Status	Target class

Target distribution

Class	Count	Share (approx.)
Stunting	696	83.2%
Overweight	94	11.2%
Underweight	43	5.1%
Wasting	4	0.5%
Total	837	100%

Imbalance interpretation

The majority-to-minority ratio is 696:4, or 174:1. A classifier that predicts Stunting for every record would already achieve about 83.2% accuracy while detecting none of the rare Wasting cases. This makes accuracy a particularly misleading headline metric.

Other supplied files

- Xtrain.csv: 1,948 rows × 4 columns; appears to be a saved training feature matrix from the whole-dataset oversampling workflow, not the original training partition.
- resampled_data.csv: 2,784 rows × 5 columns; consistent with 696 synthetic/resampled examples per class.
- svm_model.joblib, new_svm_with_grid_search.joblib and random_forest_model.joblib: serialized model artifacts. Their exact provenance and validation history are not fully documented.
- app.py references a machine-specific absolute model path. The API code is noted only as a repository artifact; deployment is outside this study pack.

Data quality questions

- What are the exact units and measurement protocols for age, height and weight?
- Are the income columns mutually exclusive one-hot indicators? What is the reference category?
- What population, age range, geography and collection period do the records represent?
- How were Status labels assigned, and are they based on accepted growth standards?
- Are there duplicate children or repeated measurements that should be grouped before splitting?